📄 Cybersecurity Project Report

Sentient Shield – SOC & EDR Grid Deployment

Project Title:
Sentient Shield – Enterprise SOC & EDR Grid Deployment

Author: Mayank
Duration: Feb 21 – Feb 27
Environment: Virtual Lab

Tools Used:

- Wazuh (SIEM / XDR)
- OpenSearch
- Ubuntu Server 22.04
- Windows Endpoint
- VirtualBox
- Sysmon
- Hydra (Attack simulation)

Executive Summary

This project implements a lightweight Security Operations Center (SOC) using Wazuh SIEM to monitor endpoint activity, detect threats, and visualize alerts. The lab environment consists of an Ubuntu-based Wazuh Manager and a Windows endpoint integrated via Wazuh Agent. Detection capabilities include log monitoring, file integrity monitoring, brute-force detection, and MITRE ATT&CK mapping. The system demonstrates the detection-analysis-response workflow used in enterprise SOC operations.

Project Objective

Goal:
Build a centralized SOC capable of:

- Endpoint monitoring
- Threat detection
- Alert visualization
- Automated response

Expected Outcome

- Windows endpoint integrated with Wazuh
- Logs flowing into dashboard
- Detection of suspicious activity
- Security events mapped to MITRE ATT&CK

Lab Architecture

Infrastructure Layout

System	Role
Kali Linux	Attacker machine
Ubuntu Server	Wazuh Manager
Windows VM	Endpoint monitored

Architecture Diagram :

Attacker (Kali)
↓
Windows Endpoint
↓
Wazuh Agent
↓
Wazuh Manager
↓
OpenSearch Indexer
↓
Dashboard

Environment Setup

Virtual Machine Configuration

Ubuntu Wazuh Manager

Setting	Value
RAM	4GB
CPU	2 cores
Disk	40GB
Network	Bridged Adapter

Week 1 – Infrastructure Deployment

Step 1 – Install Ubuntu Server

Command:

sudo apt update && sudo apt upgrade -y

Purpose:
Update system packages before installing security stack.

```
[redacted]s:~$ sudo apt update
[sudo] password for [redacted]:
Hit:1 https://brave-browser-apt-release.s3.brave.com stable InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://repository.spotify.com stable InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:6 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:7 https://ppa.launchpadcontent.net/danielrichter2007/grub-customizer/ubuntu jammy InRelease
Hit:8 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:9 https://repo.vivaldi.com/stable/deb stable InRelease
Get:10 https://repo.vivaldi.com/stable/deb stable Release [3,840 B]
Get:11 https://repo.vivaldi.com/stable/deb stable Release.gpg [833 B]
Get:12 https://repo.vivaldi.com/stable/deb stable/main amd64 Packages [1,156 B]
Hit:13 https://ppa.launchpadcontent.net/helkaluin/webp-pixbuf-loader/ubuntu jammy InRelease
Get:14 https://mega.nz/linux/repo/xUbuntu_22.04 ./ InRelease [2,961 B]
Hit:15 https://ppa.launchpadcontent.net/jonathonf/vim/ubuntu jammy InRelease
Hit:16 https://ppa.launchpadcontent.net/slimbook/slimbook/ubuntu jammy InRelease
Fetched 8,790 B in 4s (2,168 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 2 – Install Wazuh

Command:

curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh
sudo bash wazuh-install.sh -a

Result:

```
s:~$ sudo apt update
[sudo] password for :
Hit:1 https://brave-browser-apt-release.s3.brave.com stable InRelease
Hit:2 https://dl.google.com/linux/chrome/deb stable InRelease
Hit:3 http://repository.spotify.com stable InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:6 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:7 https://ppa.launchpadcontent.net/danielrichter2007/grub-customizer/ubuntu jammy InRelease
Hit:8 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Ign:9 https://repo.vivaldi.com/stable/deb stable InRelease
Get:10 https://repo.vivaldi.com/stable/deb stable Release [3,840 B]
Get:11 https://repo.vivaldi.com/stable/deb stable Release.gpg [833 B]
Get:12 https://repo.vivaldi.com/stable/deb stable/main amd64 Packages [1,156 B]
Hit:13 https://ppa.launchpadcontent.net/helkaluin/webp-pixbuf-loader/ubuntu jammy InRelease
Get:14 https://mega.nz/linux/repo/xUbuntu_22.04 ./ InRelease [2,961 B]
Hit:15 https://ppa.launchpadcontent.net/jonathonf/vim/ubuntu jammy InRelease
Hit:16 https://ppa.launchpadcontent.net/slimbook/slimbook/ubuntu jammy InRelease
Fetched 8,790 B in 4s (2,168 B/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
6 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

- **Wazuh Manager installed**
- **OpenSearch indexer configured**
- **Dashboard initialized**

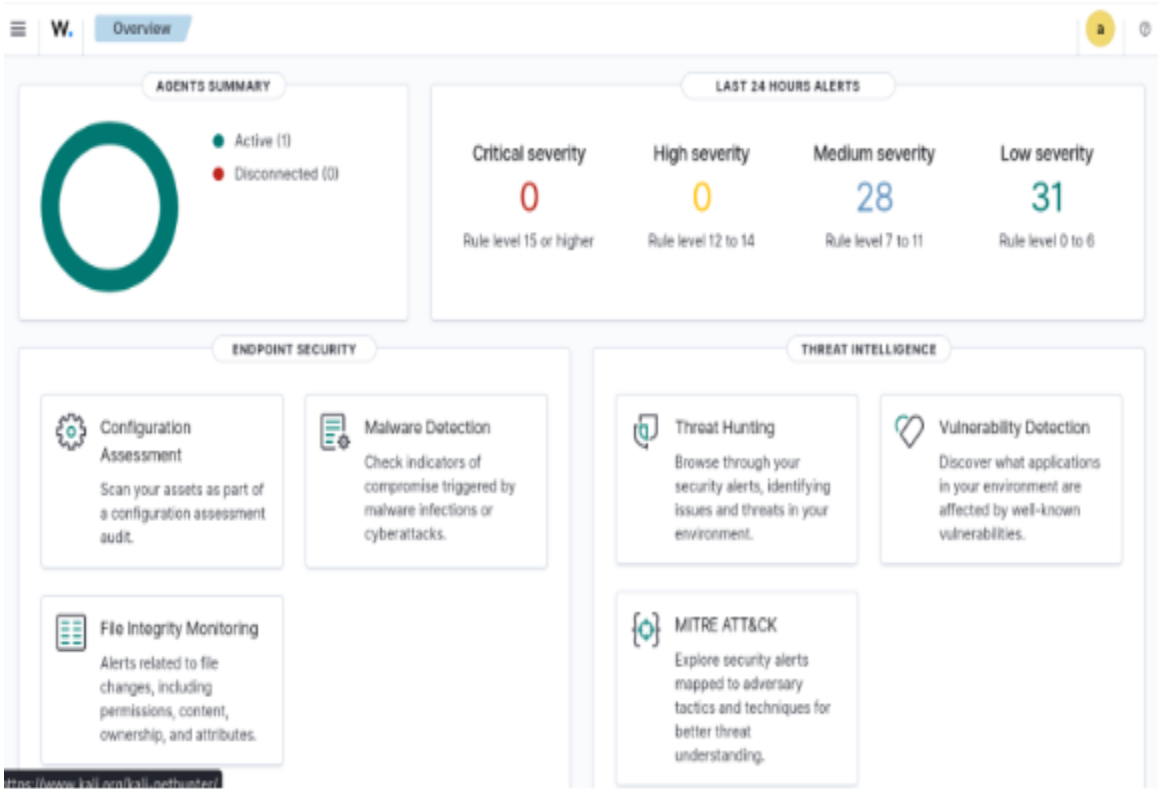
Step 3 – Access Wazuh Dashboard

Open browser:

https://<192.168.31.65>

Login using admin credentials:

Wazuh dashboard



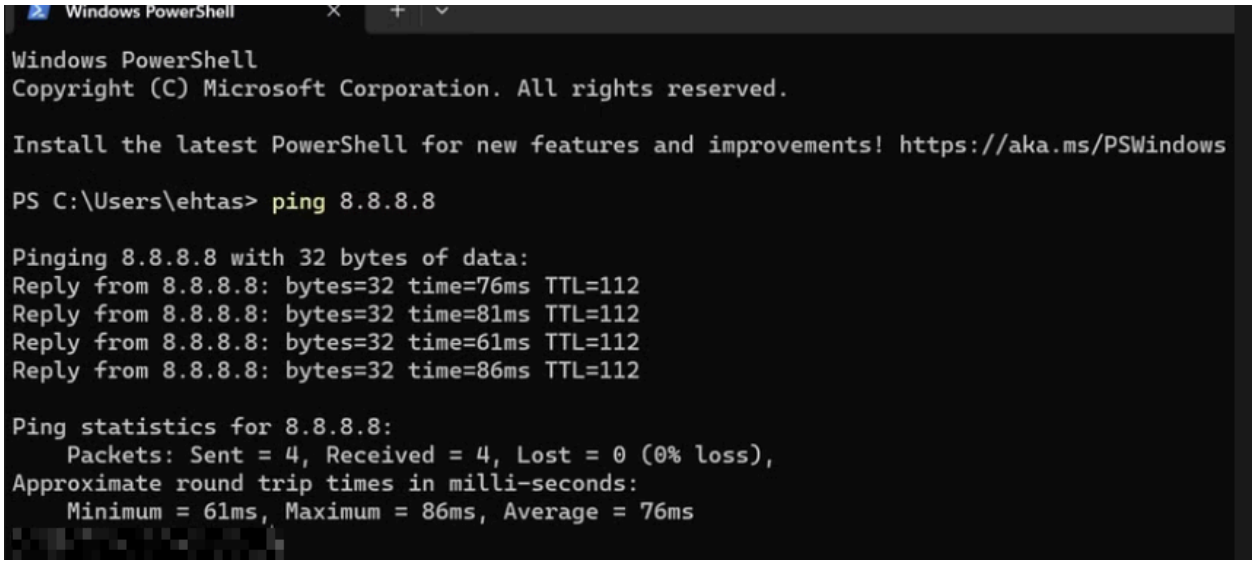
Week 2 – Endpoint Integration

Step 1 – Create Windows VM

Install Windows OS and confirm network connectivity.

Test:

ping 192.168.31.65



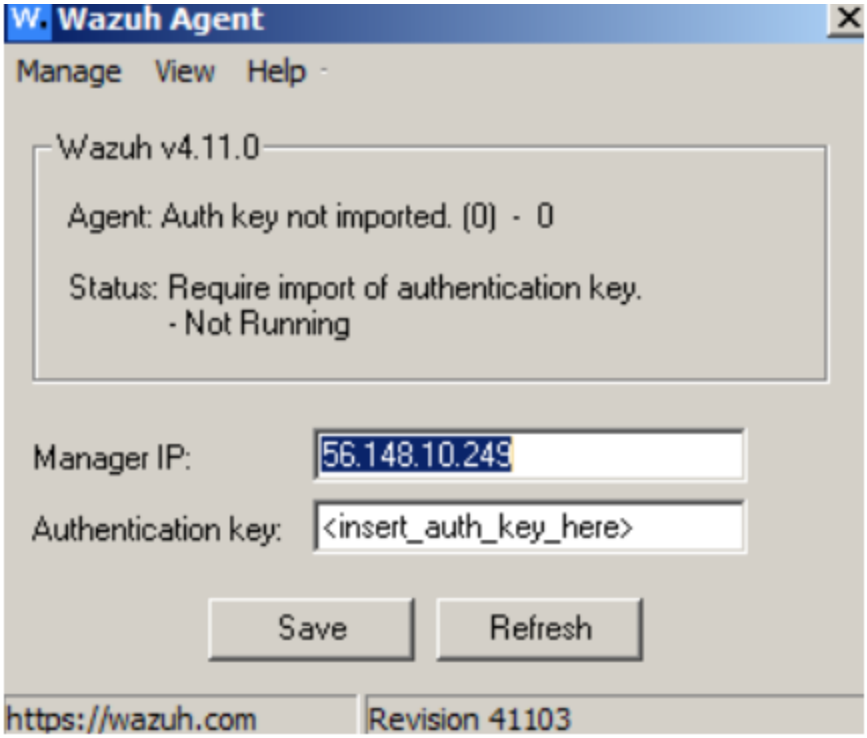
Step 2 – Install Wazuh Agent

Command (Command Prompt):

Powershell https://packages.wazuh.com/4.x/windows/wazuh-agent.msi -OutFile agent.msi

Install agent:

msiexec /i agent.msi /q WAZUH_MANAGER="192.168.31.65"



Start service:

net start wazuh

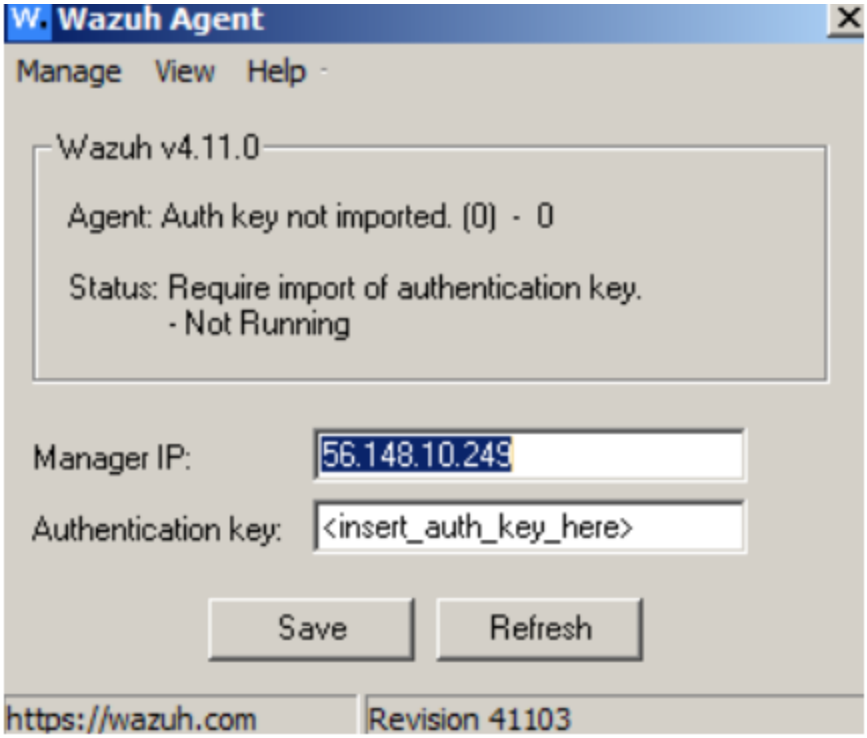
```
PS C:\Users\vagrant> NET START WazuhSvc
The requested service has already been started.

More help is available by typing NET HELPMSG 2182.
```

Step 3 – Register Agent

On Ubuntu:

sudo /var/ossec/bin/manage_agents

Add agent and extract authentication key.



Step 4 – Verify Agent Status

Command:

sudo /var/ossec/bin/agent_control -l

output:

ID: 001 | Name: Wazuh_Endpoint | Status: Active

Week 3 – Detection Engineering

File Integrity Monitoring

Monitor sensitive directories.

Example configuration:

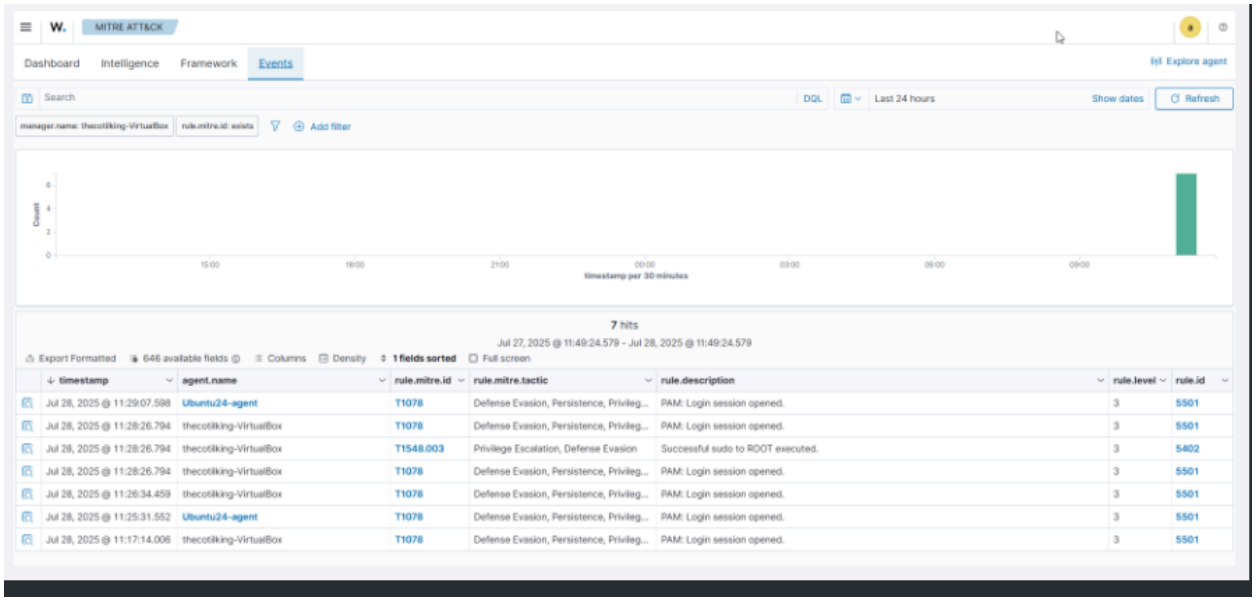
<syscheck>

<directories check_all="yes">C:\Users</directories>

</syscheck>

Purpose:

Detect unauthorized file modifications.



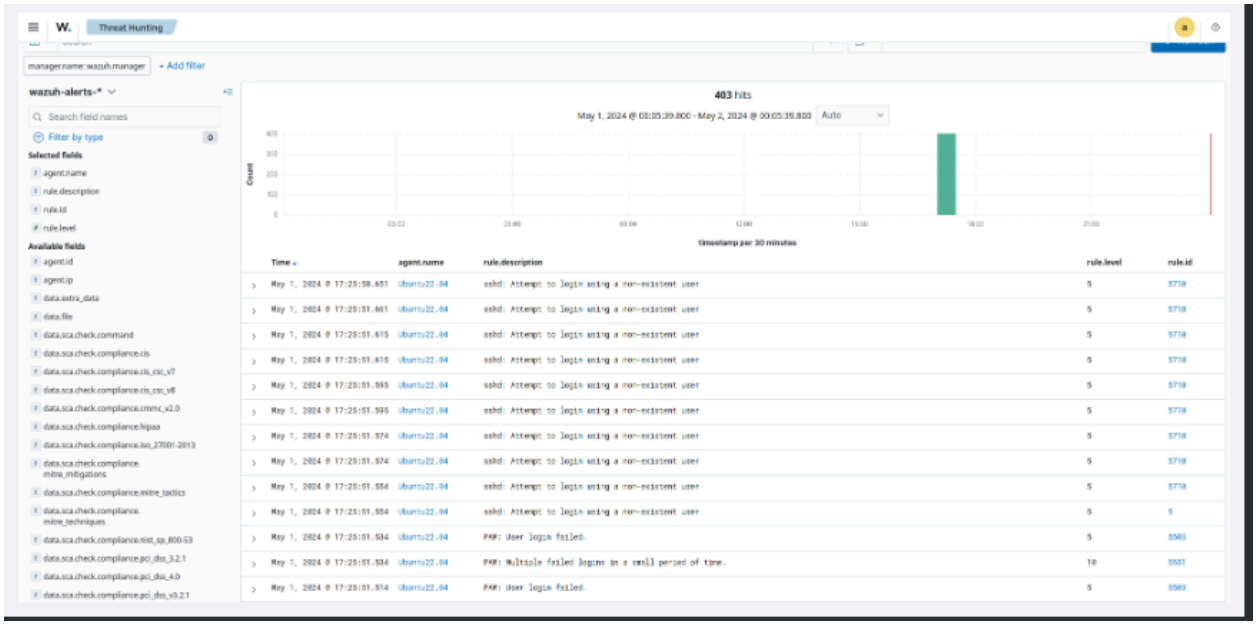
Brute Force Detection

Attack simulation from Kali:

hydra -l administrator -P rockyou.txt rdp://<target-ip>

Result:

Multiple login failures detected.



Multiple Login attempts of failed attempts suggests some one has tried to brute force login to gain excess and we are able to detect via anomaly.

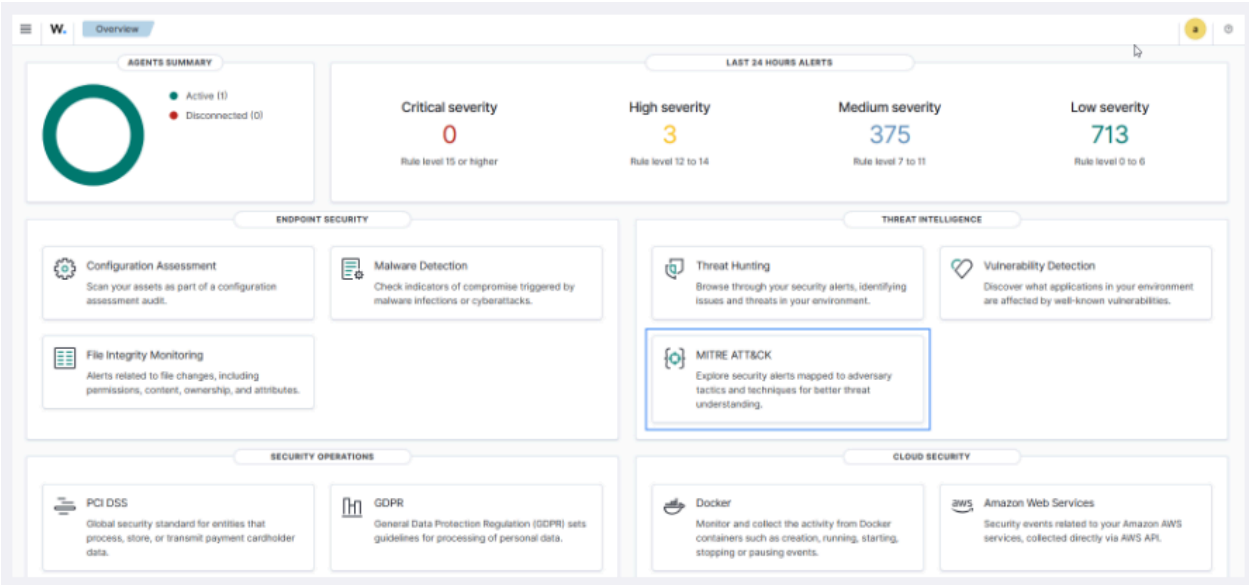
Week 4 – Threat Simulation

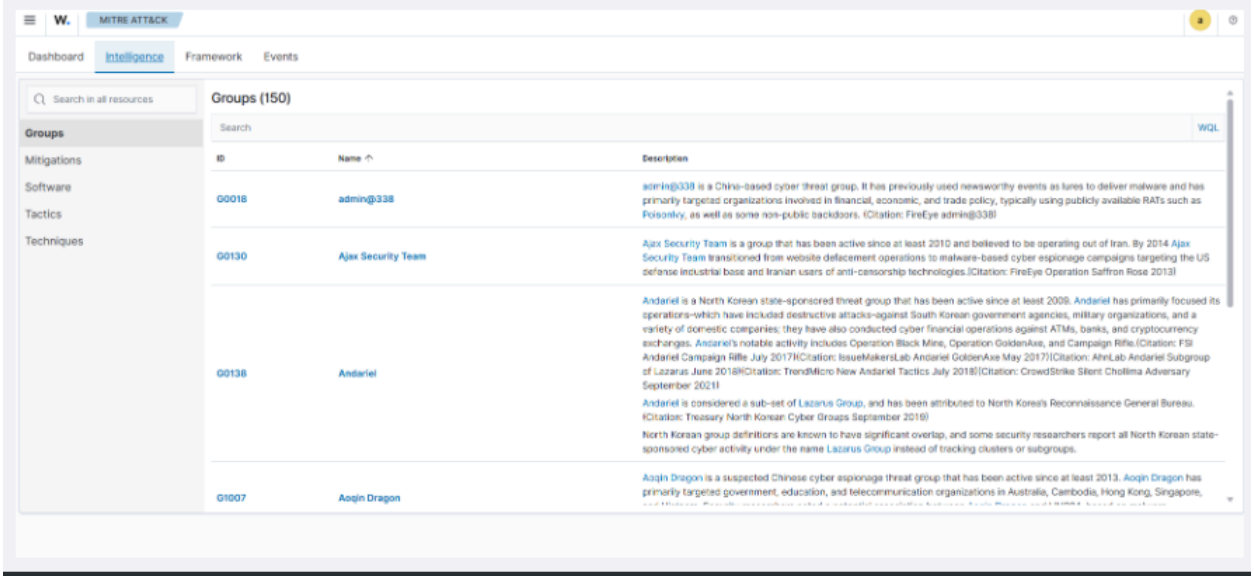
MITRE ATT&CK Mapping

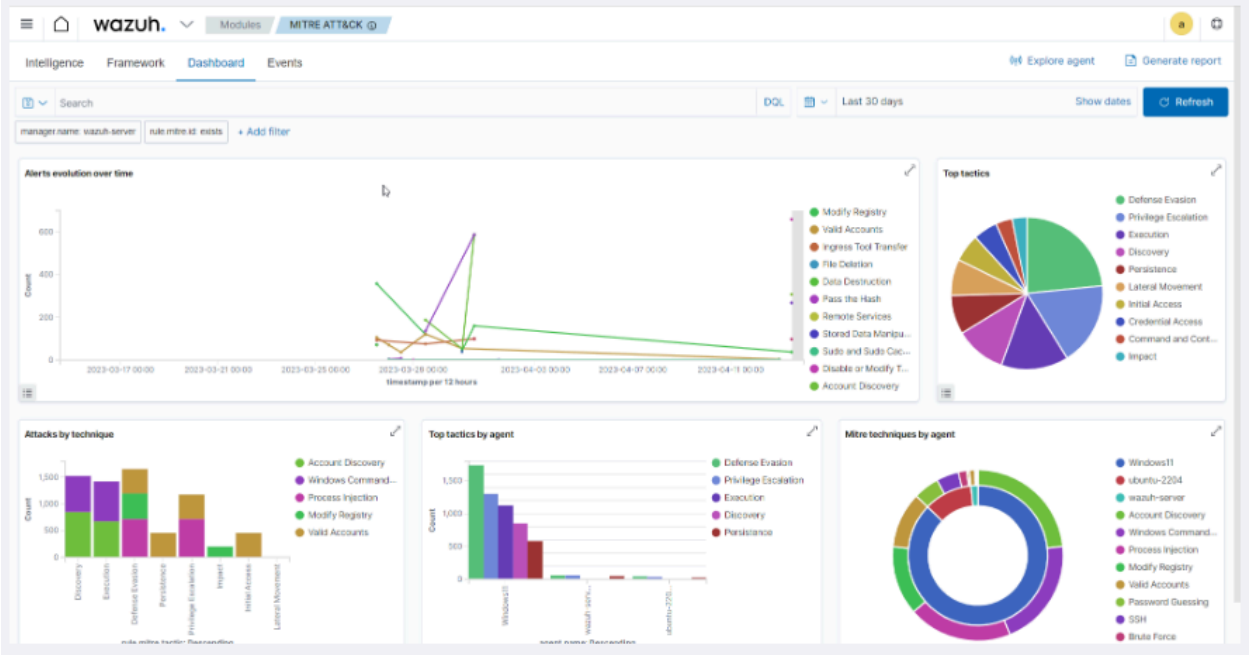
Security events mapped to attacker techniques.

Example:

Attack	MITRE Technique
Brute Force	T1110
Credential Access	T1003







Security Event Workflow

Detection logic:

Log Generated

↓

Wazuh Agent

↓

Wazuh Manager

↓

Rule Matching

↓

MITRE Mapping

↓

Alert Created

↓

Dashboard Visualization

Key Learnings

- Practical SOC architecture
- Endpoint telemetry collection
- SIEM deployment
- Threat detection workflow
- Attack simulation techniques

Challenges & Troubleshooting

Issue	Solution
Agent not connecting	Verified firewall and manager IP
Authentication errors	Re-generated agent key
IP change issue	Updated manager addres

Conclusion

The project successfully demonstrated the deployment of a lightweight SOC infrastructure using Wazuh SIEM. The system was capable of collecting endpoint logs, detecting suspicious behavior, and visualizing alerts through a centralized dashboard. This setup replicates the detection used in real-world security operations centers.